

Paper 9 — Introduction

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Introduction

Periodic elastic media offer a route to controlling wave propagation through their dispersion structure. Foundational calculations established elastic and acoustic band structures for periodic composites, and later work demonstrated complete band gaps experimentally as well as theoretically in two-dimensional solids [1, 2, 3]. Lattice analyses connected pass and stop bands to unit-cell geometry, while design studies showed how periodic architecture can be tuned for filtering and waveguiding [4, 5]. The broader principles, numerical tools, and applications of phononic materials are reviewed in the phononics literature [6, 7, 8]. These developments motivate the study of dispersion not only as a material property, but also as a means of designing directional transmission and attenuation.

A classical Cauchy continuum is a useful long-wavelength model, but its local stress–strain relation contains no intrinsic material length. When the wavelength becomes comparable to a material's microstructural scale, this scale-free description may not capture size-dependent stiffness or dispersive response. Higher-order continuum theories address this limitation by including additional kinematic measures and corresponding stress quantities. The foundations include couple-stress and microstructure theories, second-gradient elasticity, and variational treatments of the associated higher-order boundary work [9, 10, 11, 12, 13, 14]. Related size-effect research spans strain-gradient descriptions of nonuniform deformation and experiments on small-scale beams [15, 16, 17, 18, 19]. Several formulations and numerical treatments of gradient elasticity have since been developed, including unified, implicit, and finite-element approaches [20, 21, 22, 23, 24, 25]. In the present work, the constitutive setting is the local Mindlin Form-II strain-gradient model; nonlocal integral theories, couple-stress-only models, and micromorphic models are related but distinct alternatives.

Dynamic strain-gradient models require an explicit choice of kinetic energy as well as strain energy. In the model considered here, a gradient micro-inertia term supplements the classical kinetic energy. This term changes the high-wavenumber dispersion and, for the constitutive branch studied in this paper, prevents the phase-speed growth obtained when the same gradient stiffness is used with classical inertia alone. This is a model-specific admissibility issue rather than a claim that every higher-order continuum has the same high-frequency behavior. One-dimensional analyses have examined dispersion and band gaps in gradient-elastic beams and periodic media, including dipolar-gradient formulations [26, 27, 28]. At the broader generalized-continuum level, a two-dimensional phononic crystal has also been modeled with a relaxed micromorphic description containing free and gradient micro-inertia [29]. These works establish that both the continuum model and its inertia law matter when a band structure is interpreted at short wavelengths.

The phononic-crystal literature has progressively combined periodicity with size-dependent constitutive effects. Classical transfer-matrix studies of one-dimensional crystals with imperfect interfaces show that interface conditions alone can reshape the band gaps [30]. Other one-dimensional studies have considered dipolar-gradient elasticity, gradient thermoelasticity, and flexoelectric strain-gradient coupling, using analytical or transfer-matrix formulations [27, 28, 31, 32]. Two-dimensional size-dependent band structures have also been investigated: Jin, Hu, and Hu studied nano-scale crystals using nonlocal strain-gradient theory, and later examined size-dependent modes in a phononic-crystal slab [33, 34]. Related two-dimensional work includes thermoelastic cylindrical crystals and functionally graded lattices analyzed with meshless collocation or dynamic-stiffness/Floquet–Bloch methods [35, 36]. These contributions are important points of comparison, but they do not all use the same constitutive law: nonlocal strain-gradient, relaxed micromorphic, thermoelastic, flexoelectric, and local Mindlin Form-II models have different kinematics, parameters, and boundary conditions.

Anisotropy and orientation have also been studied in conventional phononic crystals. For example, calculations for orthotropic inclusions and anisotropic crystal plates show that rotating the material symmetry axes relative to the lattice can change band locations and widths [37, 38]. Such results mean that orientation-dependent band tuning itself is not new. The more specific question addressed here is whether orientation of an anisotropic characteristic-length tensor can be treated as a design variable inside a two-dimensional local strain-gradient model with gradient micro-inertia. In particular, an anisotropic length tensor derived from a rotated ellipsoidal microstructural neighbourhood is not interchangeable with rotating an orthotropic inclusion, changing a lattice direction, or introducing nonlocal interactions. This distinction is essential when comparing the present model with the classical anisotropic and nonlocal-gradient studies above.

The numerical treatment is another defining part of this problem. Bloch–Floquet conditions reduce an infinite periodic medium to a unit-cell eigenproblem, but a displacement-only strain-gradient weak form contains second spatial derivatives and therefore calls for an H^2 -conforming approximation or an equivalent mixed formulation. The bicubic Bogner–Fox–Schmit rectangle supplies C^1 inter-element continuity on structured quadrilateral meshes [39]. Earlier strain-gradient finite-element studies developed displacement-based, mixed, and implicit-gradient alternatives [40, 41, 42, 43, 44]. Subsequent work has addressed staggered and other gradient formulations, higher-order Hermite mappings, distorted domains, and mixed elements with shared or recovered degrees of freedom [22, 23, 24, 45, 46, 47, 48, 49, 50]. These methods address complementary discretization challenges; the present implementation uses a conforming C^1 BFS field so that the strain gradient is evaluated directly from the primary displacement field. For Bloch analysis of such an element, phase relations must be imposed consistently on displacement values and the derivative degrees of freedom, including the mixed derivative.

Taken together, the literature contains classical anisotropic phononic crystals, two-dimensional nonlocal strain-gradient band calculations, micromorphic models with gradient micro-inertia, and one-dimensional local gradient-elastic periodic-wave studies. To the best of our knowledge, the sources reviewed here do not combine, in one two-dimensional Bloch–Floquet finite-element framework, (i) a rotated ellipsoidal second-moment characteristic-length tensor, (ii) local Mindlin Form-II strain-gradient elasticity with an independent gradient micro-inertia, and (iii) a C^1 BFS discretization with Bloch phase transformations applied to both values and derivative degrees of freedom. The gap is therefore the combination of these model and discretization ingredients, not a general absence of anisotropy, orientation effects, or size-dependent phononic-crystal research.

The present study makes four contributions:

- C1.** A two-dimensional Bloch–Floquet framework is formulated for anisotropic strain-gradient elasticity. The characteristic-length tensor is obtained from the second moments of a rotated ellipsoidal averaging domain, so that aspect ratio $AR=l_1/l_2$ and orientation θ have distinct, physically interpretable roles.
- C2.** The role of micro-inertia is examined analytically and numerically. For the constitutive branch studied here, setting $\ell_i=0$ gives an unbounded high-wavenumber phase-velocity branch, whereas $\ell_i>0$ yields the finite limit $v_{T,\infty}=\sqrt{\mu/\rho} \ell_{\text{eff}}/(\sqrt{10} \ell_i)$ derived in Appendix A.
- C3.** A C^1 -conforming, 32-degree-of-freedom rectangular Bogner–Fox–Schmit Bloch finite element is developed. Complex Bloch phase transformations are applied consistently to nodal values and the first- and mixed-second-derivative degrees of freedom, producing a reduced generalized Hermitian eigenvalue problem.
- C4.** Direction-dependent dispersion and gap measures are evaluated over the (θ, AR) design space. The study quantifies orientation sensitivity S_θ and examines wave-vector steering, energy flux, and the influence of micro-inertia.

The remainder of the article is organized as follows. Section 2 presents the anisotropic strain-gradient continuum, ellipsoidal averaging tensor, and equations of motion with micro-inertia. Section 3 develops the Bloch–Floquet formulation and defines the reported observables. Section 4 describes the C^1 BFS element and reduced eigenproblem. Section 5 presents benchmark comparisons, internal consistency checks, and mesh convergence. Sections 6 and 7 report the dispersion, gap, orientation-sweep, wave-steering, and energy-flux results. Sections 8 and 9 discuss the findings and summarize the conclusions.

References cited in the Introduction

Numbering follows first-citation order in the current LaTeX source. The .bib file remains the machine-readable reference source.

- [1] Sigalas, M. M.; Economou, E. N. (1992). Elastic and acoustic wave band structure. *Journal of Sound and Vibration* 158(2), 377–382. [https://doi.org/10.1016/0022-460X\(92\)90059-7](https://doi.org/10.1016/0022-460X(92)90059-7)
- [2] Kushwaha, M. S.; Halevi, P.; Dobrzynski, L.; Djafari-Rouhani, B. (1993). Acoustic band structure of periodic elastic composites. *Physical Review Letters* 71(13), 2022–2025. <https://doi.org/10.1103/PhysRevLett.71.2022>
- [3] Vasseur, J. O.; Deymier, P. A.; Chenni, B.; Djafari-Rouhani, B.; Dobrzynski, L.; Prevost, D. (2001). Experimental and theoretical evidence for the existence of absolute acoustic band gaps in two-dimensional solid phononic crystals. *Physical Review Letters* 86(14), 3012–3015. <https://doi.org/10.1103/PhysRevLett.86.3012>
- [4] Martinsson, P. G.; Movchan, A. B. (2003). Vibrations of lattice structures and phononic band gaps. *The Quarterly Journal of Mechanics and Applied Mathematics* 56(1), 45–64. <https://doi.org/10.1093/qjmam/56.1.45>
- [5] Sigmund, O.; Jensen, J. S. (2003). Systematic design of phononic band-gap materials and structures by topology optimization. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences* 361(1806), 1001–1019. <https://doi.org/10.1098/rsta.2003.1177>
- [6] Hussein, M. I.; Leamy, M. J.; Ruzzene, M. (2014). Dynamics of phononic materials and structures: Historical origins, recent progress, and future outlook. *Applied Mechanics Reviews* 66(4), 040802. <https://doi.org/10.1115/1.4026911>
- [7] Maldovan, M. (2013). Sound and heat revolutions in phononics. *Nature* 503(7475), 209–217. <https://doi.org/10.1038/nature12608>
- [8] Khelif, A.; Adibi, A. (2015). *Phononic Crystals: Fundamentals and Applications*. Springer New York. <https://doi.org/10.1007/978-1-4614-9393-8>
- [9] Toupin, R. A. (1962). Elastic materials with couple-stresses. *Archive for Rational Mechanics and Analysis* 11(1), 385–414. <https://doi.org/10.1007/BF00253945>
- [10] Mindlin, R. D.; Tiersten, H. F. (1962). Effects of couple-stresses in linear elasticity. *Archive for Rational Mechanics and Analysis* 11, 415–448. <https://doi.org/10.1007/BF00253946>
- [11] Mindlin, R. D. (1964). Micro-structure in linear elasticity. *Archive for Rational Mechanics and Analysis* 16(1), 51–78. <https://doi.org/10.1007/BF00248490>
- [12] Mindlin, R. D. (1965). Second gradient of strain and surface-tension in linear elasticity. *International Journal of Solids and Structures* 1(4), 417–438. [https://doi.org/10.1016/0020-7683\(65\)90006-5](https://doi.org/10.1016/0020-7683(65)90006-5)
- [13] Mindlin, R. D.; Eshel, N. N. (1968). On first strain-gradient theories in linear elasticity. *International Journal of Solids and Structures* 4(1), 109–124. [https://doi.org/10.1016/0020-7683\(68\)90036-X](https://doi.org/10.1016/0020-7683(68)90036-X)
- [14] Germain, P. (1973). The Method of Virtual Power in Continuum Mechanics. Part 2: Microstructure. *SIAM Journal on Applied Mathematics* 25(3), 556–575. <https://doi.org/10.1137/0125053>
- [15] Aifantis, E. C. (1999). Strain gradient interpretation of size effects. *International Journal of Fracture* 95, 299–314. <https://doi.org/10.1023/A:1018625006804>
- [16] Fleck, N. A.; Hutchinson, J. W. (1993). A phenomenological theory for strain gradient effects in plasticity. *Journal of the Mechanics and Physics of Solids* 41(12), 1825–1857. [https://doi.org/10.1016/0022-5096\(93\)90072-N](https://doi.org/10.1016/0022-5096(93)90072-N)
- [17] Fleck, N. A.; Hutchinson, J. W. (1997). Strain gradient plasticity. *Advances in Applied Mechanics* 33, 295–361. [https://doi.org/10.1016/S0065-2156\(08\)70388-0](https://doi.org/10.1016/S0065-2156(08)70388-0)
- [18] Yang, F.; Chong, A. C. M.; Lam, D. C. C.; Tong, P. (2002). Couple stress based strain gradient theory for elasticity. *International Journal of Solids and Structures* 39(10), 2731–2743. [https://doi.org/10.1016/S0020-7683\(02\)00152-X](https://doi.org/10.1016/S0020-7683(02)00152-X)
- [19] Lam, D. C. C.; Yang, F.; Chong, A. C. M.; Wang, J.; Tong, P. (2003). Experiments and theory in strain gradient elasticity. *Journal of the Mechanics and Physics of Solids* 51(8), 1477–1508. [https://doi.org/10.1016/S0022-5096\(03\)00053-X](https://doi.org/10.1016/S0022-5096(03)00053-X)
- [20] Askes, H.; Aifantis, E. C. (2002). Numerical modeling of size effects with gradient elasticity—formulation, meshless discretization and examples. *International Journal of Fracture* 117(4), 347–358. <https://doi.org/10.1023/A:1022225526483>
- [21] Askes, H.; Aifantis, E. C. (2011). Gradient elasticity in statics and dynamics: An overview of formulations, length scale identification and applications. *International Journal of Solids and Structures* 48(13), 1968–1990. <https://doi.org/10.1016/j.ijsolstr.2011.03.006>
- [22] Askes, H.; Aifantis, E. C. (2006). Gradient elasticity theories in statics and dynamics—a unification of approaches. *International Journal of Fracture* 139(2), 297–304. <https://doi.org/10.1007/s10704-006-8375-4>
- [23] Askes, H.; Gutiérrez, M. A. (2006). Implicit gradient elasticity. *International Journal for Numerical Methods in Engineering* 67(3), 400–416. <https://doi.org/10.1002/nme.1640>
- [24] Askes, H.; Morata, I.; Aifantis, E. C. (2008). Finite element analysis with staggered gradient elasticity. *Computers & Structures* 86, 1266–1279. <https://doi.org/10.1016/j.compstruc.2007.11.002>
- [25] Polyzos, D.; Fotiadis, D. I. (2012). Derivation of Mindlin's first and second strain gradient elastic theory via simple lattice models. *International Journal of Solids and Structures* 49(3–4), 470–480. <https://doi.org/10.1016/j.ijsolstr.2011.10.021>
- [26] Papargyri-Beskou, S.; Beskos, D. E. (2009). Dynamic analysis of gradient elastic flexural beams. *International Journal of Solids and Structures* 46(10), 2151–2159. <https://doi.org/10.1016/j.ijsolstr.2009.01.024>

- [27] Li, Y. Q.; Wei, P. J.; Zhou, Y. (2016). Band gaps of elastic waves in 1-D phononic crystal with dipolar gradient elasticity. *Acta Mechanica* 227(4), 1005–1023. <https://doi.org/10.1007/s00707-015-1495-z>
- [28] Hosseini, S. M.; Sladek, J.; Sladek, V.; Zhang, C. (2024). Effects of the strain gradients on the band structures of the elastic waves propagating in 1D phononic crystals: An analytical approach. *Thin-Walled Structures* 194, 111316. <https://doi.org/10.1016/j.tws.2023.111316>
- [29] Madeo, A.; Collet, M.; Miniaci, M.; Billon, K.; Ouisse, M.; Neff, P. (2018). Modeling phononic crystals via the weighted relaxed micromorphic model with free and gradient micro-inertia. *Journal of Elasticity* 130(1), 59–83. <https://doi.org/10.1007/s10659-017-9633-6>
- [30] Zheng, M.; Wei, P. J. (2009). Band gaps of elastic waves in 1-D phononic crystals with imperfect interfaces. *International Journal of Minerals, Metallurgy and Materials* 16(5), 608–614. [https://doi.org/10.1016/S1674-4799\(09\)60105-9](https://doi.org/10.1016/S1674-4799(09)60105-9)
- [31] Li, Y.; Askes, H.; Gitman, I. M.; Krynkin, A.; Wei, P. (2023). Band gaps of thermoelastic waves in 1D phononic crystal with fractional order generalized thermoelasticity and dipolar gradient elasticity. *Waves in Random and Complex Media* 36(4), 5715–5735. <https://doi.org/10.1080/17455030.2023.2222189>
- [32] Li, Y.; Li, Y.; Guo, Z.; Wang, H.; Wang, C. (2024). Band gaps of elastic waves in 1-D dielectric phononic crystal with the flexoelectric and strain gradient effects consideration. *Scientific Reports* 14, 24035. <https://doi.org/10.1038/s41598-024-75049-1>
- [33] Jin, J.; Hu, N.; Hu, H. (2022). Investigation of size effect on band structure of 2D nano-scale phononic crystal based on nonlocal strain gradient theory. *International Journal of Mechanical Sciences* 219, 107100. <https://doi.org/10.1016/j.ijmecsci.2022.107100>
- [34] Jin, J.; Hu, N.; Hu, H. (2023). Size effects on the mixed modes and defect modes for a nano-scale phononic crystal slab. *Applied Mathematics and Mechanics (English Edition)* 44(1), 21–34. <https://doi.org/10.1007/s10483-023-2945-6>
- [35] Hosseini, S. M.; Zhang, C. (2021). Band structure analysis of Green-Naghdi-based thermoelastic wave propagation in cylindrical phononic crystals with energy dissipation using a meshless collocation method. *International Journal of Mechanical Sciences* 209, 106711. <https://doi.org/10.1016/j.ijmecsci.2021.106711>
- [36] Mishra, M.; Kumar, S.; Sharma, A. (2026). Tunable wave propagation and band gap characteristics in functionally graded lattices with multiple topologies via dynamic stiffness and Floquet–Bloch formulation. *Acta Mechanica* 237, 3951–3982. <https://doi.org/10.1007/s00707-026-04680-y>
- [37] Zhan, Z. Q.; Wei, P. J. (2010). Influences of anisotropy on band gaps of 2D phononic crystal. *Acta Mechanica Solida Sinica* 23(2), 181–188. [https://doi.org/10.1016/S0894-9166\(10\)60020-1](https://doi.org/10.1016/S0894-9166(10)60020-1)
- [38] Yao, Y.; Wu, F.; Hou, Z.; Xin, Z. (2011). Lamb waves in two-dimensional phononic crystal plate with anisotropic inclusions. *Ultrasonics* 51(5), 602–605. <https://doi.org/10.1016/j.ultras.2010.12.016>
- [39] Bogner, F. K.; Fox, R. L.; Schmit, L. A. (1965). The generation of inter-element-compatible stiffness and mass matrices by the use of interpolation formulas. *Proceedings of the Conference on Matrix Methods in Structural Mechanics* 397–443
- [40] Shu, J. Y.; King, W. E.; Fleck, N. A. (1999). Finite elements for materials with strain gradient effects. *International Journal for Numerical Methods in Engineering* 44(3), 373–391. [https://doi.org/10.1002/\(SICI\)1097-0207\(19990130\)44:3<373::AID-NME508>3.0.CO;2-7](https://doi.org/10.1002/(SICI)1097-0207(19990130)44:3<373::AID-NME508>3.0.CO;2-7)
- [41] Amanatidou, E.; Aravas, N. (2002). Mixed finite element formulations of strain-gradient elasticity problems. *Computer Methods in Applied Mechanics and Engineering* 191(15–16), 1723–1751. [https://doi.org/10.1016/S0045-7825\(01\)00353-X](https://doi.org/10.1016/S0045-7825(01)00353-X)
- [42] Zervos, A.; Papanastasiou, P.; Vardoulakis, I. (2001). A finite element displacement formulation for gradient elastoplasticity. *International Journal for Numerical Methods in Engineering* 50(6), 1369–1388. [https://doi.org/10.1002/1097-0207\(20010228\)50:6<1369::AID-NME72>3.0.CO;2-K](https://doi.org/10.1002/1097-0207(20010228)50:6<1369::AID-NME72>3.0.CO;2-K)
- [43] Zervos, A.; Papanicolopulos, S.-A.; Vardoulakis, I. (2009). Two finite-element discretizations for gradient elasticity. *Journal of Engineering Mechanics* 135(3), 203–213. [https://doi.org/10.1061/\(ASCE\)0733-9399\(2009\)135:3\(203\)](https://doi.org/10.1061/(ASCE)0733-9399(2009)135:3(203))
- [44] Beheshti, Alireza (2017). Finite element analysis of plane strain solids in strain-gradient elasticity. *Acta Mechanica* 228(10), 3543–3559. <https://doi.org/10.1007/s00707-017-1897-1>
- [45] Andreaus, U.; dell'Isola, F.; Giorgio, I.; Placidi, L.; Lekszycki, T.; Rizzi, N. L. (2016). Numerical simulations of classical problems in two-dimensional (non) linear second gradient elasticity. *International Journal of Engineering Science* 108, 34–50. <https://doi.org/10.1016/j.ijengsci.2016.08.003>
- [46] Choi, Jae-Hoon; Lee, Byung-Chai; Sim, Gi-Dong (2023). Mixed finite elements based on superconvergent patch recovery for strain gradient theory. *Computer Methods in Applied Mechanics and Engineering* 411, 116053. <https://doi.org/10.1016/j.cma.2023.116053>
- [47] Papanicolopulos, Stefanos-Aldo (2024). Efficient strain-gradient mixed elements using shared degrees of freedom for the discretised fields. *International Journal for Numerical Methods in Engineering* 125(16), e7536. <https://doi.org/10.1002/nme.7536>
- [48] Chirkov, Aleksandr Yu.; Nazarenko, Lidiia; Altenbach, Holm (2024). Plane crack problems within strain gradient elasticity and mixed finite element implementation. *Computational Mechanics* 74(3), 703–721. <https://doi.org/10.1007/s00466-024-02451-x>
- [49] Baccocchi, Michele; Fantuzzi, Nicholas (2023). On the mapping procedure based on higher-order Hermite polynomials for laminated thin plates with arbitrary domains in gradient elasticity. *International Journal for Numerical Methods in Engineering* 124(11), 2552–2570. <https://doi.org/10.1002/nme.7220>
- [50] Baccocchi, Michele; Fantuzzi, Nicholas (2025). Free vibrations and bending of laminated thin plates with distorted domains in gradient elasticity: an isoparametric finite element formulation based on Hermite mapping. *Acta Mechanica*. <https://doi.org/10.1007/s00707-025-04254-4>